POST-QUANTUM READINESS REPORT

protected.example

18 October 2026



Quantum ready

Connections to this host are protected against future quantum decryption.

THE ONE NEXT STEP

Retire RSA key transport cipher suites

PUBLIC RESULT. Observed from the public Internet; contains no private data and may be shared. Live result: https://quantumreadylabs.com/r/demo

EXECUTIVE **SUMMARY**

On 18 October 2026, Quantum Ready Labs tested protected.example (203.0.113.10) from the public Internet: fourteen TLS probes and one ordinary connection, the same traffic a browser visit causes. The question was a narrow one: where does this host stand on the post-quantum transition, and what is the next step.

Connections to this host are protected against future quantum decryption. The key exchange, which is the urgent half of the transition, accepts the hybrid post-quantum group that current browsers already send; traffic recorded today cannot be decrypted later by a quantum computer.

Authentication (the certificate chain) is reported but not scored against the host: no certificate on the public web can be made quantum-safe yet, and a signature cannot be forged retroactively, so there is nothing to do there today and nothing to fear. The stage name is the summary; the score under it (85 of 100) is for those who want a number.

THE ONE NEXT STEP

Retire RSA key transport cipher suites

TLS 1.2 sessions that use RSA key transport have no forward secrecy: one recorded session falls when the RSA key falls, with or without a quantum computer. Keep TLS 1.2 for old clients if you must, but only with ECDHE suites.



READINESS

EXPOSED
TLS 1.3 is not available: sessions can be recorded today and decrypted later.

AT RISK
TLS 1.3 with forward secrecy, no post-quantum key exchange: safe today, harvestable.

PROTECTED
Hybrid post-quantum key exchange is accepted: confidentiality protected.

READY
Protected, with no legacy key transport and short-lived automated certificates.

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WHERE THE HOST **STANDS**

EXPOSED
TLS 1.3 is not available: sessions can be recorded today and decrypted later.

AT RISK
TLS 1.3 with forward secrecy, no post-quantum key exchange: safe today, harvestable.

THIS HOST

PROTECTED
Hybrid post-quantum key exchange is accepted: confidentiality protected.

READY
Protected, with no legacy key transport and short-lived automated certificates.

KEY EXCHANGE · CONFIDENTIALITY

Hybrid post-quantum accepted

The urgent half: traffic recorded today with a classical key exchange can be decrypted when a large quantum computer exists.

- Hybrid groups: X25519MLKEM768
- Classical groups: x25519, secp256r1, secp384r1, secp521r1
- Server's pick for a current browser: X25519MLKEM768
- TLS 1.3: available
- TLS 1.2 RSA key transport: still accepted (no forward secrecy)

CERTIFICATE · AUTHENTICATION

Classical, like all of the public web

Nothing to do today, and nothing to fear: the Web PKI has no post-quantum roots yet, and a signature cannot be forged after the fact.

- Key: P-256
- Signature: ecdsa-with-SHA256
- Lifetime: 90 days, 47 remaining
- Chain: 4 links; trusted: yes

FOUNDATIONS · AGILITY

4 of 5 in place

The habits that make the next transition cheap: modern protocol, no legacy fallbacks, automated short-lived certificates.

- TLS 1.3: yes
- No RSA key transport: no, still accepted
- Short-lived certificate (100 days or less): yes
- Server prefers hybrid when offered: yes
- Chain trusted: yes

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TECHNICAL **DETAILS**

PROBES

PROBE	ANSWER
X25519MLKEM768 Hybrid post-quantum (standard)	Supported
SecP256r1MLKEM768 Hybrid post-quantum (standard)	Not supported
SecP384r1MLKEM1024 Hybrid post-quantum (standard)	Not supported
X25519Kyber768Draft00 Hybrid post-quantum (pre-standard draft)	Supported
x25519 Classical elliptic curve	Supported
secp256r1 Classical elliptic curve	Supported
secp384r1 Classical elliptic curve	Supported
secp521r1 Classical elliptic curve	Supported
ffdhe2048 Classical finite field	Not supported
ffdhe3072 Classical finite field	Not supported
TLS 1.3 Protocol	Available
TLS 1.2 with ephemeral keys Protocol, forward secrecy	Negotiated
TLS 1.2 RSA key transport Legacy, no forward secrecy	Still accepted
Server's pick for a current browser Offered what Chrome offers	X25519MLKEM768

CERTIFICATE CHAIN

LINK	SUBJECT	ISSUER	KEY	SIGNATURE	UNTIL
Leaf	protected.example	WE1	P-256	ecdsa-with-SHA256	2026-12-04
Intermediate	WE1	GTS Root R4	P-256	ecdsa-with-SHA384	2029-02-20
Intermediate	GTS Root R4	GlobalSign Root CA	P-384	sha256WithRSAEncryption	2028-01-28
Root	GlobalSign Root CA	GlobalSign Root CA	RSA-2048	sha1WithRSAEncryption	2028-01-28

CONNECTION

A real connection negotiated TLSv1.3 with TLS_AES_128_GCM_SHA256, ALPN h2. Scan time 105 ms from 203.0.113.10.

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METHOD, RUBRIC AND **STANDARDS**

HOW THE HOST WAS TESTED

The engine builds TLS ClientHello messages by hand and reads the server's first reply in clear: offered one key-exchange group with no key share, a supporting server answers HelloRetryRequest naming it; a non-supporting server sends an alert. No post-quantum cryptography is implemented, no handshake completes, and the result is identical on every platform. One ordinary connection reads the negotiated parameters and the certificate chain. Fourteen probes, run concurrently, on port 443.

WHY KEY EXCHANGE, AND NOT THE CERTIFICATE

A quantum computer able to break RSA and elliptic curves does not exist. The threat today is "harvest now, decrypt later": encrypted traffic recorded now falls when such a machine exists. That is solved in the key exchange, which current browsers, OpenSSL 3.5 and the large CDNs already do with hybrid post-quantum groups. Certificates authenticate the moment a connection happens; a signature cannot be forged retroactively, and the Web PKI has no post-quantum roots yet.

SCORE 85 OF 100

Key exchange	60 / 60
No RSA key transport	0 / 10
Short-lived certificate	8 / 8
Server prefers hybrid	6 / 6
Trusted chain	6 / 6
Certificate algorithms	5 / 10

THE STAGES

EXPOSED	TLS 1.3 is not available: sessions can be recorded today and decrypted later.
AT RISK	TLS 1.3 with forward secrecy, no post-quantum key exchange: safe today, harvestable.
PROTECTED	Hybrid post-quantum key exchange is accepted: confidentiality protected.
READY	Protected, with no legacy key transport and short-lived automated certificates.

STANDARDS

NIST FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), FIPS 205 (SLH-DSA); NIST IR 8547 (quantum-vulnerable algorithms deprecated after 2030, disallowed after 2035); NSA CNSA 2.0; IETF draft-ietf-tls-ecdhe-mlkem (the hybrid groups). Full rubric: quantumreadylabs.com, How we score.

RE-TEST

Results are a snapshot. Re-test after any change at quantumreadylabs.com; this result stays at https://quantumreadylabs.com/r/demo.

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